Name: _____

Class: _____

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Embedded Systems: Key Facts and Activity

Key information

Key concept

An embedded system is a computer system built into a larger device to do one dedicated job.

Typical characteristics:

- it does **one specific job**, and only that job;
- its program is **firmware**, stored in **ROM**;
- its processor is **small, cheap and low-power**, because the job is simple;
- it has **simple input and output**: buttons and sensors in; motors, lights or a display out.

Key concept

Inside the box. Most embedded systems are built around a **microcontroller**: a CPU, a fixed amount of RAM and a fixed amount of ROM, all on **one circuit board**. The manufacturer stores the firmware in the ROM at the factory, and the user never interacts with it directly.

Why use one? Because the job is fixed, it is efficient, reliable, simple to design, cheap, compact and low in power use.

Every embedded system follows one pattern: input, process, output. Sensors and buttons send **inputs**; the firmware **processes** them by following its rules; motors, lights, speakers and screens carry out the **outputs**.

Worked example

Worked example: a video doorbell.
Input: the button, the camera and the motion sensor. **Process:** the firmware spots a press or a movement and decides to raise an alert. **Output:** the chime in the house, and a notification with video on the owner's phone.

Embedded or general-purpose? Two questions decide it: *does it do one dedicated job?* and *is it built into a larger device?* Two yeses means embedded.

	Embedded system	**General-purpose computer**
Job	one dedicated job	many different jobs
Program	firmware in ROM	software on secondary storage, loaded into RAM
Processor	small, cheap, low-power	powerful

Where you find them.

In the home	smart meter, video doorbell, burglar alarm, smart lighting
On the body	smart watch, fitness tracker, heart-rate monitor, earbuds
In a vehicle	anti-lock brakes, cruise control, climate control, airbag control unit
On the street	traffic lights, speed camera, CCTV, air-quality monitor

Microcontroller and benefits points follow the Isaac Computer Science page "Embedded systems", in our own words. Further reading: 101computing.net, "Embedded Systems in a Smart World".

Part 1 Be the firmware: a pedestrian crossing

How it works. Work in a group of four. Together you are the embedded system inside a pedestrian crossing. Each person takes one role and keeps it for all three scenarios.

Role	Your job	What you may do
Button (input)	Press when a pedestrian arrives.	Say "Pressed". Nothing else.
Timer (input)	Count the seconds on the class clock.	Say "5 seconds", "8 seconds" and so on, when that time has passed.
Firmware (processor)	Read the rules below. Pick the rule that matches the inputs you have heard.	Use only the rules on this page. Give one command at a time, such as "Car lights amber".
Lights (output)	Carry out each command and show the state.	Hold up the colour on your fingers: 1 = green, 2 = amber, 3 = red.

Key concept

The firmware: its rules are stored in ROM.

Rule 1 At the start: car lights green, pedestrian light red.
Rule 2 IF the button is pressed, THEN wait 5 seconds, then car lights amber for 2 seconds, then car lights red.
Rule 3 WHEN the car lights are red, THEN pedestrian light green and beeper on for 8 seconds.
Rule 4 WHEN the 8 seconds are over, THEN beeper off, pedestrian light red, car lights green.

Run the three scenarios. The timekeeper writes in the table: the input heard, the rule the firmware chose, and the output the lights showed. *Scenario 1:* one pedestrian presses the button. *Scenario 2:* the button is pressed twice during the 5-second wait. *Scenario 3:* nobody presses the button for 30 seconds.

Scen.	Input heard	Rule used	Output the lights showed
1			
1			
1			
2			
3			

Check your result. In every scenario the firmware used *only* the four rules, and the lights matched the rule it chose. Scenario 1 ends where it began: car green, pedestrian red.

Swap roles and run Scenario 1 again. Everyone has now been an input, the processor and an output. Timekeeper: write the new total time for one crossing. . . . . seconds

Part 2 Embedded, or general-purpose?

Apply the two questions from page 1: *Does it do one dedicated job?* and *is it built into a larger device?* Two yeses means embedded. Tick one column for each device.

Device	Embedded	General-purpose	Why (one word)
Automatic hand dryer			
Gaming laptop			
Electric toothbrush timer			
Tablet for reading and games			
Car engine control unit			
School file server			
Heart-rate monitor on a wristband			
Home desktop PC			

Check your result. Four devices are embedded and four are general-purpose. Each general-purpose device can have new programs installed; each embedded one runs the one program its maker stored.

Part 3 Say it in exam language

Use what the crossing showed you. Complete each sentence.

1. The crossing does . . . . . and cannot run a game or a web browser. That is why it is an . . . . . and not a general-purpose computer.

2. The four rules are the crossing's . . . . . . It is stored in . . . . . , so it is still there when the power comes back on, because ROM is . . . . . .

3. The crossing needs only a . . . . . processor, because following four rules is a simple job.

4. Name the **input** devices and the **output** devices in the real crossing.

Inputs: . . . . . Outputs: . . . . . .

Stretch. The council wants the crossing to stay green for pedestrians for 12 seconds instead of 8. **(a)** Which rule changes? . . . . . .

(b) The firmware is in ROM. Explain how the change could still be made, and why the council does not do it often.

. . . . . .

. . . . . .

Part 4 Going further: embedded systems in a smart world

Complete the table for each device, as the video doorbell was done on page 1. Write one input, the process in a short phrase, and one output.

Device	Input	Process	Output
Fitness tracker			
Car tyre-pressure monitor			
Smart streetlight			

Check your result. Each row names a sensor or button as the input, and a light, screen, lamp or speaker as the output. The process in between is a decision the firmware makes.

(a) A microcontroller holds a CPU, ... and . . . . . on one circuit board.

(b) Give **two** benefits of using an embedded system in a streetlight rather than a general-purpose computer.

. . . . . .

. . . . . .

What does "smart" add? A smart device connects to the internet, using Wi-Fi, Bluetooth or mobile data. The video doorbell is one example.

(c) Describe what the connection lets the video doorbell do that an ordinary doorbell cannot.

. . . . . .

. . . . . .

(d) The video doorbell is still an embedded system. Complete the sentence.

It is embedded because it does . . . . . , which is watching the door, and it is built into the . . . . . .